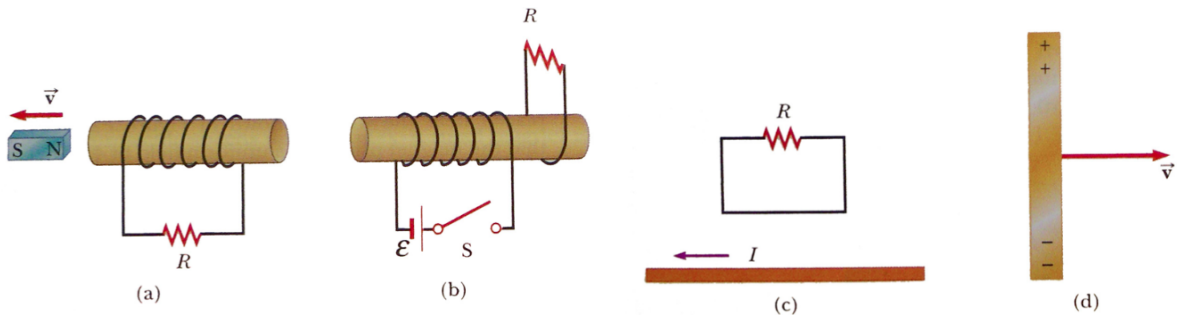


Physik II — FS 2024 — Prof. R. Wallny — Serie 11

Release: 14th May 2024 – Due date and discussion: 23rd May 2024

Exercise 1: Lenz's rule

Objective: Lenz's rule, right-hand-rule, Lorentz force.



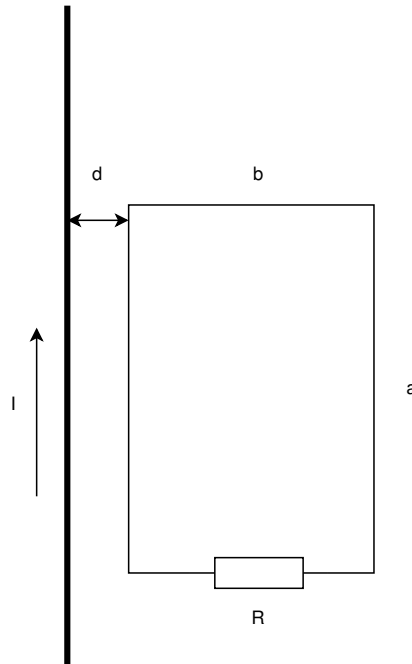
Use Lenz's rule to answer the following questions about the pictured circuits:

- What direction does the current induced in resistor R assume, if the magnet moves to the left?
- What direction does the current induced in resistor R assume, shortly after the switch has been closed?
- What direction does the current induced in resistor R assume, if the current I in the figure suddenly drops to zero?
- A copper bar is moved perpendicular to a magnetic field towards the right. If it is charged as indicated in the figure, what would be the direction of the magnetic field?

Exercise 2: Conductor loop at High Voltage line

Objective: Induction in changing magnetic field

A rectangular, closed conductor loop with side lengths $b = 15\text{ cm}$ and $a = 20\text{ cm}$ is located at a distance of $d = 1\text{ cm}$ next to a high-voltage line as shown in the sketch. The high-voltage line carries a direct current (HVDC) of 2400 A . The intrinsic resistance in the conductor loop is $R = 1.2 \cdot 10^{-4}\ \Omega$.

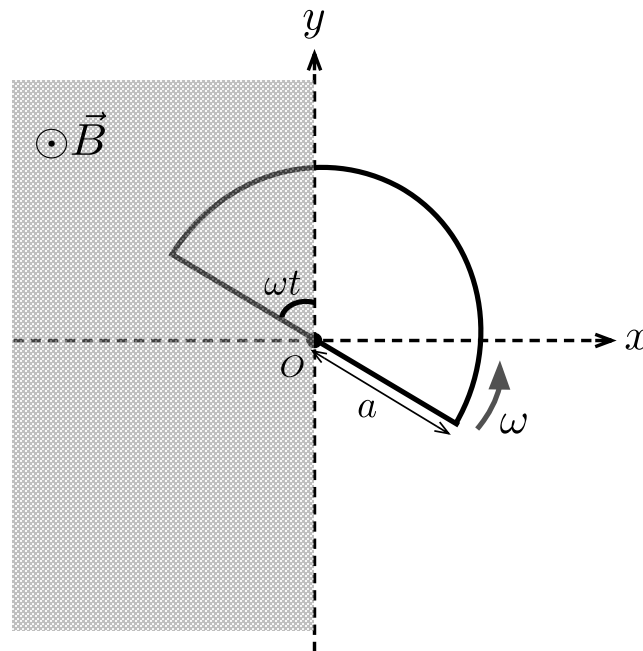


- What is the magnetic field strength B at distances d and $d + b$ from the high-voltage line?
- Calculate the magnetic flux Φ through the conductor loop.
The current in the high-voltage line is now being interrupted. The current within the line decreases at a constant rate to 0 A within a time $\Delta t = 0.4\text{ ms}$.
- How strong is the induced current in the conductor loop?
- During the turning off of the high voltage line, a resulting force acts on the conductor loop. In which direction does it act and how strong is it?

Exercise 3: Semicircle generator (14.5 Points)

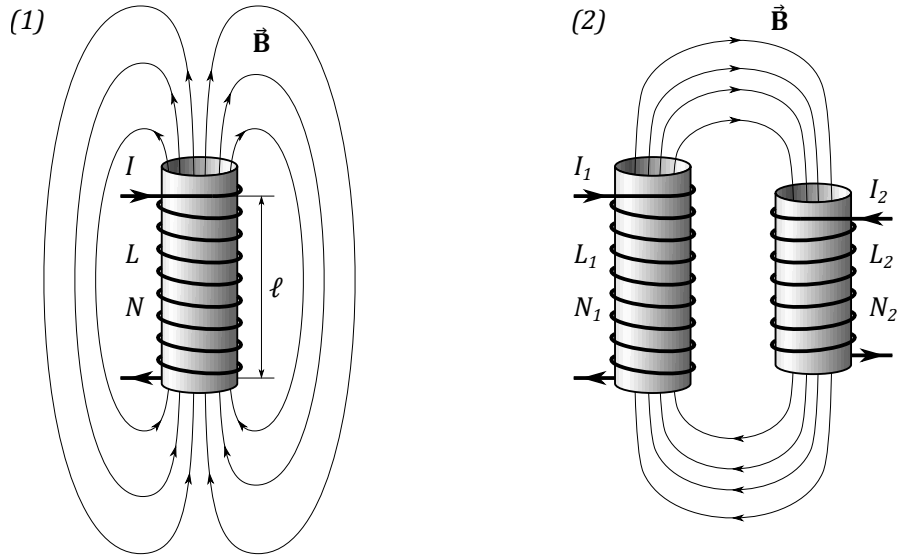
The half-space $x < 0$ of a Cartesian coordinate system has a uniform magnetic field $\vec{B} = B\vec{e}_z$, while the field for $x > 0$ is zero (see figure below). A semicircular closed loop with radius a and resistance R lies in the xy -plane, with its center at the origin O of the coordinate system. The loop rotates with a constant angular velocity ω counterclockwise around the z -axis. At time $t = 0$ the complete loop surface is located in the $x \geq 0$ half of the xy -plane.

- Calculate the magnetic flux through the coil for times $t \in [0; T/2]$, where T denotes one rotational period.
- In which direction (clockwise, counterclockwise) does the current flow for times $t \in [0; T/2]$? (T is the period of the rotation)? In which direction for $t \in [T/2; T]$? Draw the time evolution of the current I neglecting the self induction.
- Calculate the peak value of the induced current in the loop .
- Evaluate the external torque that needs to be exerted to keep the coil rotating at constant angular velocity ω .
- What external mechanical power is needed to keep the loop rotating at constant angular velocity?



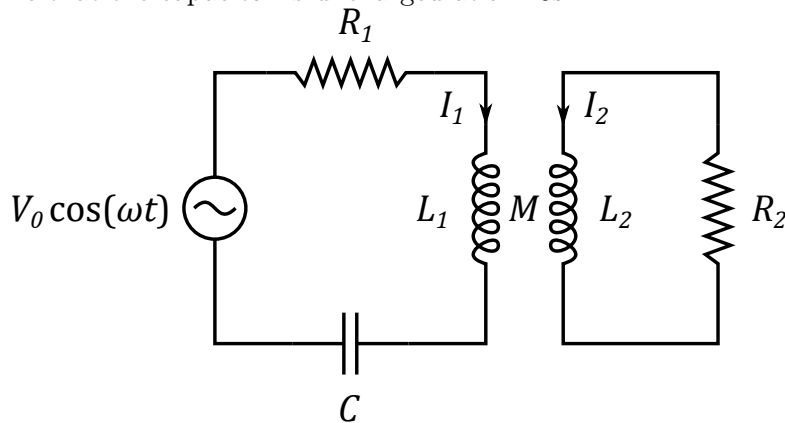
Exercise 4: Mutual inductance

Objective: self-inductance, mutual inductance



- a) Consider a solenoid coil of length ℓ and N windings of radius R , as shown in figure (1). Assume that the B-field inside the coil is given by $B = \mu_0 \frac{N}{\ell} I$. Starting from $L = \phi/I$, deduce the self-inductance L (ϕ the N -fold magnetic flux through the coil). Express L as a function of N , R and ℓ .
- b) Express the B-field inside the coil as a function of L , R , N and I .
- c) Consider two coils with the same radius R , but different inductances L_1 , L_2 and numbers of windings N_1 and N_2 , as shown in figure (2). The field produced by one coil is transported into the other coil without loss¹. Consequently, the field B_{21} , which is produced by coil 1 inside coil 2, is equal to the field B_{11} , which is produced by coil 1 inside coil 1. Calculate the mutual inductance M of this system as a function of L_1 and L_2 .
- d) The two coils from figure (2) have been incorporated into the circuit outlined below. Write down the two differential equations for I_1 and I_2 .

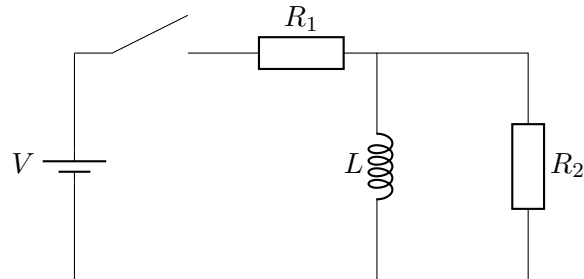
Hint: Assume that the capacitor is uncharged at $t = 0$ s.



¹Technically, this is accomplished with an iron core. For simplicity, we neglect this iron core.

Exercise 5: Energy in an inductor

Objective: circuit with R- and L-components, Kirchhoff's laws, solution of differential equations with initial conditions, in a resistor dissipated energy, in an inductor stored magnetic energy.



In the shown circuit the resistance of L is negligible and the switch is open, therefore initially the current is zero.

- a) Find the energy dissipated as heat in the resistance R_2 when the switch is closed and remains closed for a long time.
Hint: Determine $I_L(t)$ first, using the initial conditions, and then determine the current through R_2 ($I_2(t)$) to get the dissipated power.
- b) Find now the energy dissipated in R_2 when the switch, after being closed for a long time, is opened and remains open for a long time.

Use the values $V = 100\text{ V}$, $R_1 = 10\text{ Ohm}$, $R_2 = 100\text{ Ohm}$ and $L = 10\text{ H}$, but apply them as late as possible.